Answers

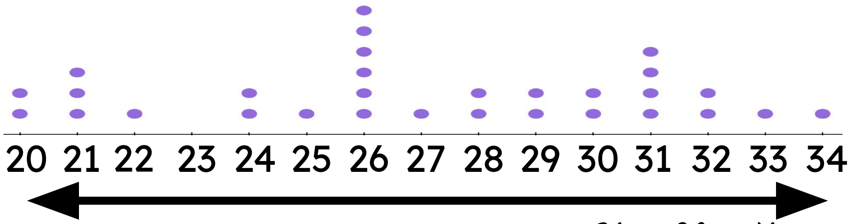
Worksheet

Understanding and calculating the range

Task A: What is the range?

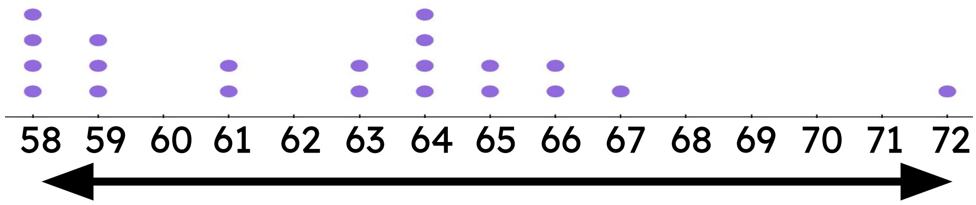
1) For each dot plot, work out the **range**.

a)



range = 34 − 20 = 14

b)



range = 72 − 58 = 14

2) For each set of data, work out the **range**.

a) −1, 0, 1, 1, 1, 2, 4, 5, 7, 7, 8, 8, 8, 10, 10, 10, 14, 17, 17, 20

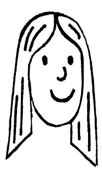
range = 20 − (−1) = 21

b) −7, 6, 6, −3, −9, −9, −5, 6, −1, 6, 5, −6, −8, −4, −4, −3, 4, −6, 2, 6

range = 6 − (−9) = 15

c) 16.2, 15.3, 1.1, 19.4, 13.7, 6.7, 4.7, 12.1, 16.3, 9.4, 8.4, 18.2, 11.9, 1.6, 17.8

range = 19.4 − 1.1 = 18.3

3) 

Laura had just finished her homework on working out the **range** when her ink pen started to leak. What number has the ink spoiled in each case?

a) Largest number is *24*, smallest number is 14 so the **range** is 10

b) Largest number is 8, smallest number is *-4* so the **range** is 12

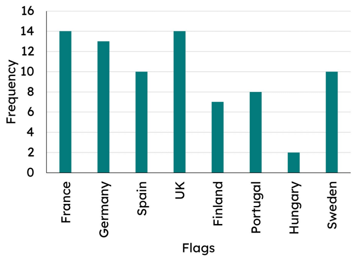
c) In order: −3, −1, −1, 0, *e.g. 2*, 6, 7, 10 so the **range** is 13 — *Any number between 0 and 6*

d) Any order: 3, 5, 3, 7, 10, 12, 6, 8, *e.g. 1* so the **range** is 11 — *1 or 14*

Task B: Extreme values in different representations

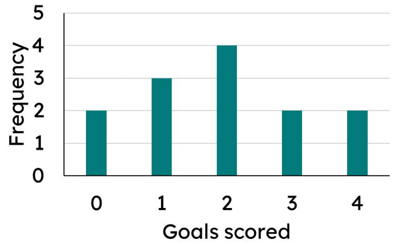
1) Which of these representations can you find the **range?**

a)



✗

b)



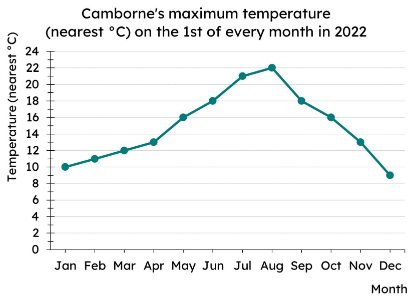
✓

c)

Shoe size	3	4	5	6
Frequency	7	12	10	3

✓

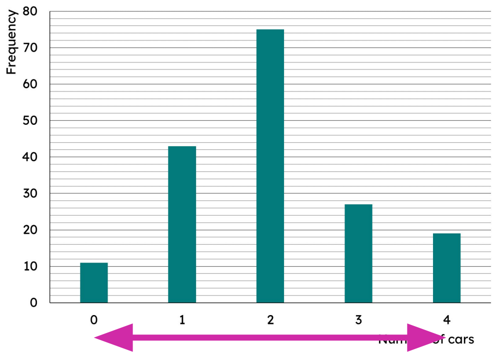
d)



✓

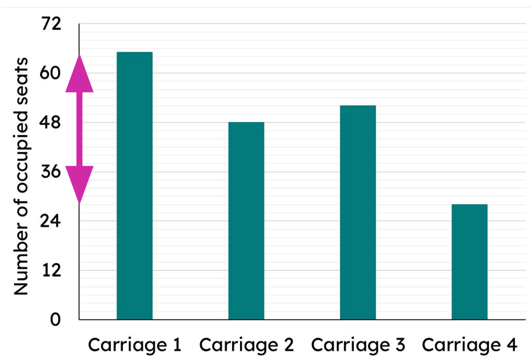
2) For each of the following bar charts, find the **range.**

a)



Range of cars: 4 – 0 = 4

b)



Range of occupied seats: 65 – 28 = 37

3) Some data is represented in this frequency table about wait times at a fast food restaurant.

Wait time (mins)	4	5	6	7	8	9	10	11
Frequency	46	20	17	13	16	10	8	8

A local newspaper reported that customers' wait time **ranged** by 38 minutes. Do you agree?

*No, the **range** of wait times is 7 mins; from the longest wait time of 11 minutes to the shortest wait time of 4 minutes.*